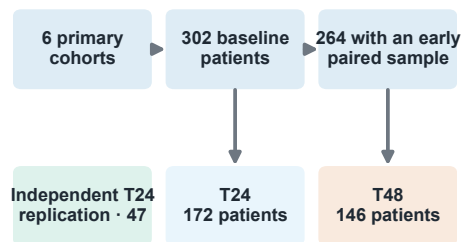


A Longitudinal cohort architecture



Clinical context: GSE95233, GSE110487, GSE57065 and GSE54514

B Evidence layers

Temporal drift

mean within-patient ΔZ

Rank persistence

baseline-to-follow-up ordering

Clinical context

source-defined course or severity

Independent replication

direction and transcriptome-matched null

Formula architecture

exact gene-level decomposition

C Eight fixed published formulas

Sepsis / infection

MetaScore

difference of geometric means

SeptiCyte

four-gene linear contrast

FAIM3:PLAC8

two-gene ratio

sNIP

three-gene normalized contrast

Pathogen class

B/V MetaScore

difference of geometric means

Herberg DRS

two-gene linear contrast

Outcome / severity

Lin score

seven-gene weighted sum

Severe/Mild

four-module ratio

Published gene membership and algebra were retained; no formula was refitted in the longitudinal cohorts.